FIITJEE

ALL INDIA TEST SERIES

FULL TEST –XI

JEE (Advanced)-2020

PAPER – 2

TEST DATE: 23-09-2020

ANSWERS, HINTS & SOLUTIONS

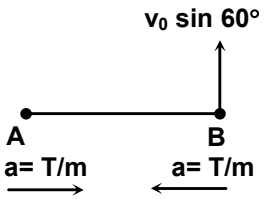
Physics

PART – I

SECTION – A

1. D

Sol. Looking at B with respect to A : $2T = \frac{m(v_0 \sin 60°)^2}{\ell}$



2. A, D

Sol. For ring: $(\mu Nr)dt = I\omega_0$ (Rotation)

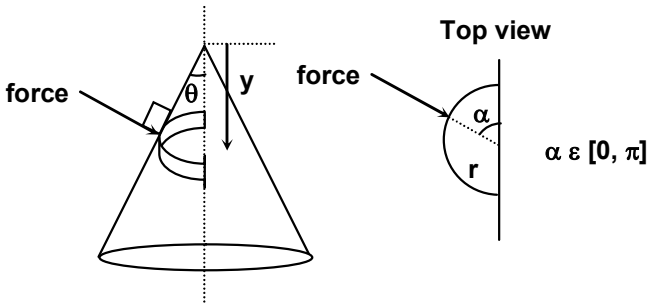
$(N\cos\theta + \mu N \sin\theta)dt = m\sqrt{2gH}$ (translation)

For wedge : $(N \sin\theta)\, dt = (\mu\, N \cos\theta)dt$

3. B

Sol. Net force on the element at depth y,

$dF(y) = \rho gy\left(\frac{dy}{\cos\theta} \times rd\alpha \times \cos\theta \times \sin\alpha\right)$



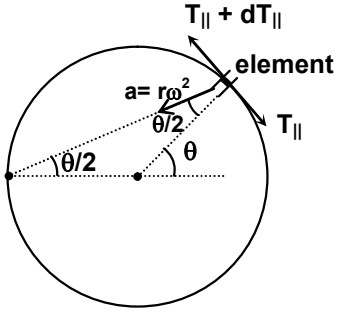
$$\therefore\ F = \int_0^H dF(y)$$

and effective point of application = $\dfrac{\int_0^H y\,dF(y)}{\int_0^H dF(y)}$

4. B

Sol. $dT_{||} = (dm)r\omega^2 \sin\dfrac{\theta}{2}$

$$\Rightarrow\ T = T_A + \frac{MR\omega^2}{2\pi}(1-\cos\theta)$$



5. A, C, D

Sol. $\Delta Q_{AB} = nC_P\Delta T = 2\times\dfrac{5R}{2}(900-300) = 3000R$

In the process BC, $P = \alpha T^2$

PV^2 = constant

Molar heat capacity in the process BC,

$$C = C_V + \frac{R}{(1-x)} = \frac{3R}{2} + \frac{R}{(1-2)}$$

$$C = \frac{R}{2}$$

$$\Delta Q_{BC} = nC\Delta T = 2\times\frac{R}{2}(300-900) = -600R$$

$$\Delta Q_{CA} = \Delta W_{CA} = nRT\ell n\left(\frac{P_C}{P_A}\right) = 2R\times 300\ell n\left(\frac{1}{9}\right) = -1200R\ell n(3)$$

$$\Delta Q_{CA} = -1320R$$

$$\Delta W_{cycle} = \Delta Q_{cycle} = 3000R - 600R - 1320R = 1080R$$

Efficiency of cycle, $\eta = \dfrac{\Delta W_{cycle}}{\Delta Q_{supplied}}\times 100$

$$\eta = \frac{1080R}{3000R}\times 100 = 36\%$$

6. A, D

Sol. $\vec{B}$ due to body diagonal $= \vec{0}$

$\vec{B}$ due to edge $\frac{\mu_0 i}{4\pi(a/\sqrt{2})}\left(\sqrt{\frac{1}{3}}+\sqrt{\frac{1}{3}}\right)$along$\frac{-\hat{i}-\hat{j}}{\sqrt{2}}$ (using the result $\frac{\mu_0 i}{4\pi r}(\sin\alpha+\sin\beta)$

$\vec{B}$ due to face diagonal = $\frac{\mu_0 i}{4\pi(a/2)}\left(\sqrt{\frac{2}{3}}+\sqrt{\frac{2}{3}}\right)$along $\frac{-\hat{i}-\hat{j}}{\sqrt{2}}$

Also, magnetic moment $\vec{\mu}=i\vec{A}$.

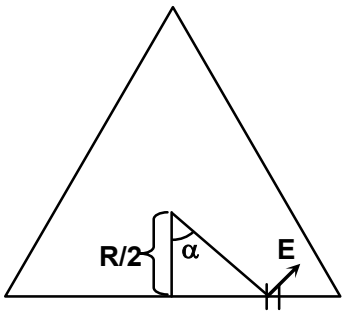
7. B, C, D

Sol. The motion will be helical. Also, at each collision, the motion will be mirror image of the motion that would have happened without collision (mirror being the wall).

8. A, C

Sol. $E_{induced} = \frac{B_0 r}{2}$

Net torque $= 6\int_0^{60^0}\lambda\left(\frac{R}{2}\sec^2\alpha d\alpha\right)\times E\times\frac{R/2}{\cos\alpha}=I\alpha$



9. B

10. B

Sol. **(for Q.9-Q.10):**

For any case, Buoyancy $\vec{B}=-V\rho_{liquid}\vec{g}_{eff}$

where $\vec{g}_{eff}=\vec{g}-\vec{a}$ and $\vec{T}+\vec{B}+m\vec{g}=m\vec{a}$

11. D

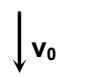
Sol. Find time constant by shorting the battery finding R_{eff} across C_0. Then $\tau = R_{eff}\times C_0$.

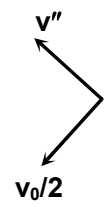
12. A

Sol. Find time constant by shorting the battery finding C_{eff} across R_0. Then $\tau = R_0\times C_{eff}$.

SECTION – C

13. 00001.69

Sol. Let A :  & B : $\leftarrow v_0$ before collision

Let A :  & B : $\leftarrow v'$ after collision

$e = 1 \Rightarrow v_0\left(\frac{1}{2}+\frac{\sqrt{3}}{2}\right) = v'' - \frac{v'}{2}$ . . . (i)

Conserving momentum horizontally $\Rightarrow mv_0 = mv' + m\left(\frac{v''}{2}+\frac{v_0\sqrt{3}}{4}\right)$ . . . (ii)

Solve for v'.

14. 00002.30

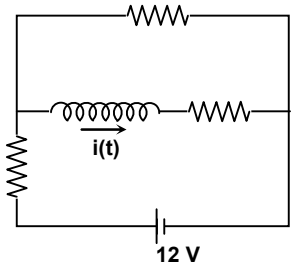
Sol. $i(t) = i_{max}(1 - e^{-t/\tau})$

here, $i_{max} = V/3R$

$\tau = 2L/3R$

$\Rightarrow i_{bulb}(t) = \frac{V}{6R}(2 + e^{-3Rt/2L})$

$\Rightarrow \Delta t = \frac{2L}{3R}\ln 2$.



15. 00500.00

Sol. $X_L = \omega L = 100 \times 0.1 = 10\ \Omega$

$\frac{1}{Z'} = \frac{1}{R_2} + \frac{1}{X_L i} = \frac{1}{10} + \frac{1}{10i} = \frac{1-i}{10}$

$Z' = \frac{10}{(1-i)} = \frac{10}{2}(1+i) = 5(1+i)$

$Z' = 5(1+i)$

Net impedance of the circuit, $|Z| = \frac{250}{10} = 25\,\Omega$

$Z = (R_1 - X_C i) + Z' = (15 - X_C i) + 5(1+i)$

$Z = 20 + (5 - X_C)i$

$|5 - X_C| = \sqrt{(25)^2 - (20)^2}$

$|5 - X_C| = 15$

$X_C - 5 = 15$

$X_C = 20\ \Omega$

$\Rightarrow \frac{1}{\omega C} = 20$

$\Rightarrow C = \frac{1}{20\omega} = \frac{1}{20 \times 100} = 500\mu F$

$C = 500\ \mu F$

16. 00004.70

Sol. $\frac{13.6Z^2}{n^2} = 36\%$ of 13.6 . . . (i)

Also, ${}^nC_2 = 10$ . . . (ii)

and, $eV_0 = \frac{hc}{\lambda} - \phi$ . . . (iii)

17. 00000.40

Sol. $W(t) = N_0e^{-4\lambda t}$ . . . (i)

$$\frac{dZ}{dt} = +\lambda W(t) \quad \text{. . . (ii)}$$

$$\frac{dX}{dt} = +3\lambda W(t) - 2\lambda X(t) \quad \text{. . . (iii)}$$

Solve the three equations.

18. 00003.14

Sol. After passing through prism, ray from upper prism becomes horizontal & ray from lower prism is inclined at 2^0 to the horizontal.

$\Rightarrow$ $\Delta x = d \sin 1^0 + d \sin \theta$ . . . (i)

Also, $I = 4I_0 \cos^2 \left(\frac{\Delta\phi}{2}\right) = \frac{1}{2}(4I_0) = 2I_0$

$\Rightarrow$ $\cos^2 \left(\frac{\Delta\phi}{2}\right) = \frac{1}{2}$ . . . (ii)

Solve (1) & (2).

Chemistry

PART – II

SECTION – A

19. B, C, D

Sol. $Cu + \text{conc.}HNO_3 \longrightarrow \underset{\text{'K'}}{Cu(NO_3)_2} + \underset{\text{'L'}}{NO_2} \uparrow + \underset{\text{'M'}}{H_2O}$

$Cu(NO_3)_2 + KCN(\text{excess}) \longrightarrow \underset{\text{'T'}}{K_3[Cu(CN)_4]} + \underset{\text{'S'}}{(CN)_2} \uparrow + KNO_3$

20. A, B, D

Sol. $Al_2O_3.xH_2O + \underset{\text{'Q'}}{NaOH} \longrightarrow \underset{\text{'R'}}{Na[Al(OH)_4]}$

$Na[Al(OH)_4] + CO_2 \longrightarrow Al_2O_3.xH_2O \downarrow + NaHCO_3$

'R' 'S' 'T'

21. A, B

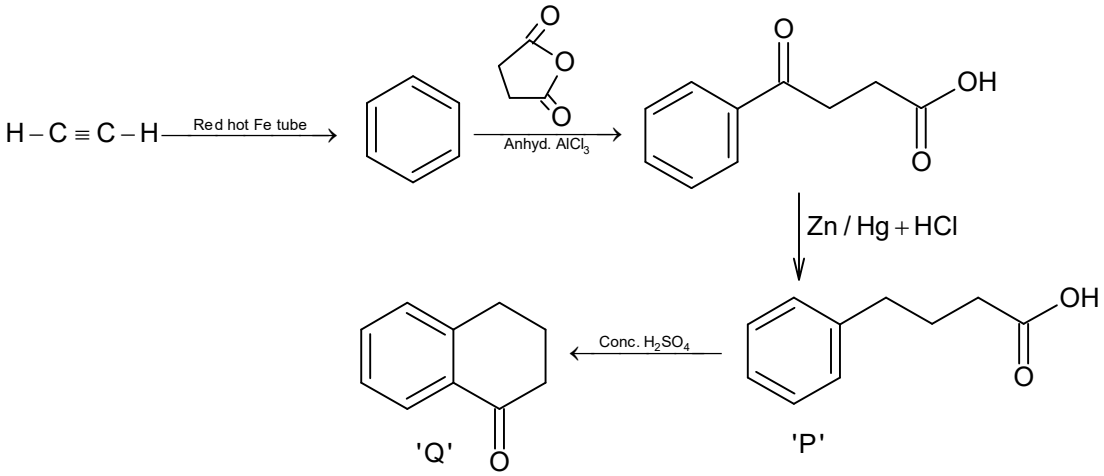
Sol. $\underset{\text{Excess}}{3Cl_2} + NH_3 \longrightarrow NCl_3 + 3HCl$

$Cl_2 + NaOH \xrightarrow[\text{hot conc.}]{} NaCl + NaClO_3 + H_2O$

22. C, D

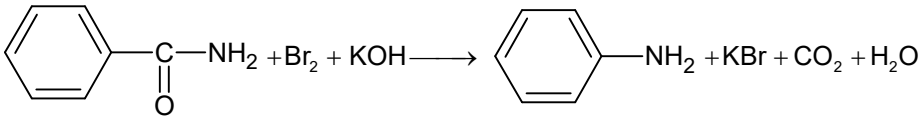
23. C, D

Sol.



24. B, C, D

Sol.



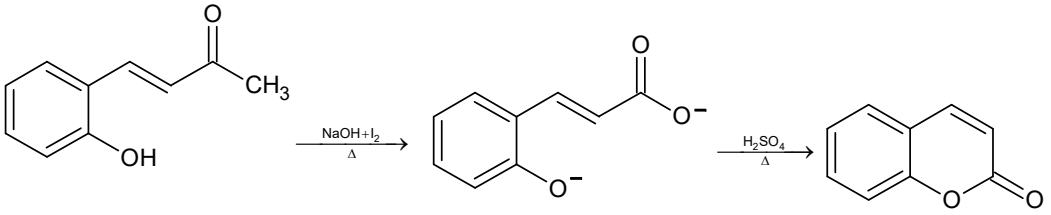
25. C, D

26. B, C

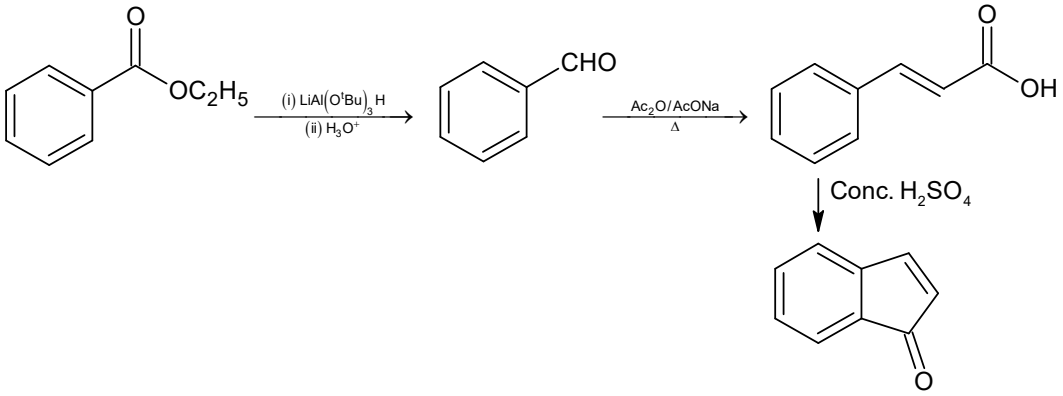
27. D

28. C

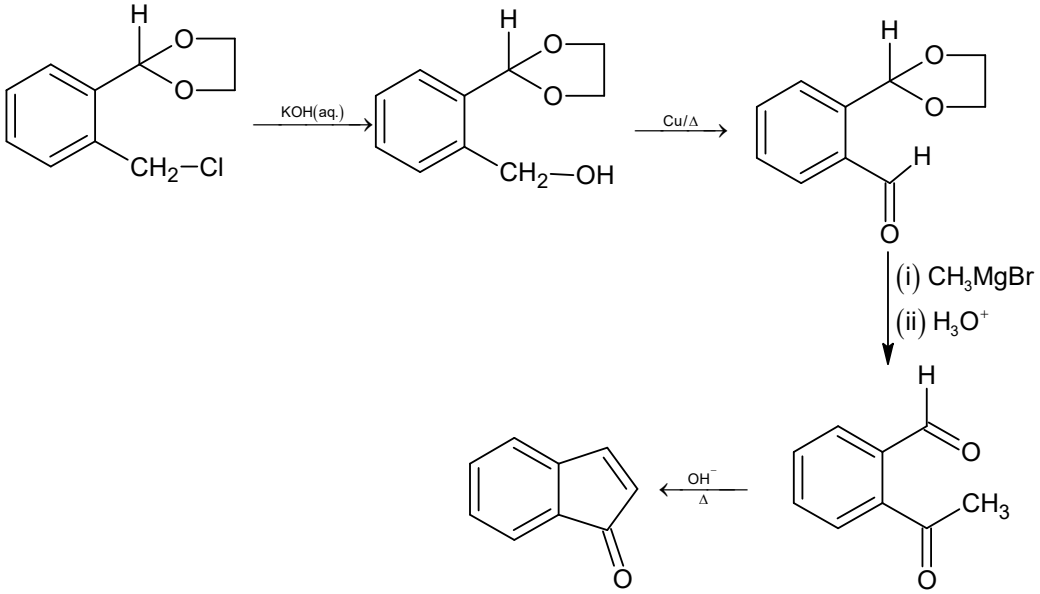
Sol. (I)



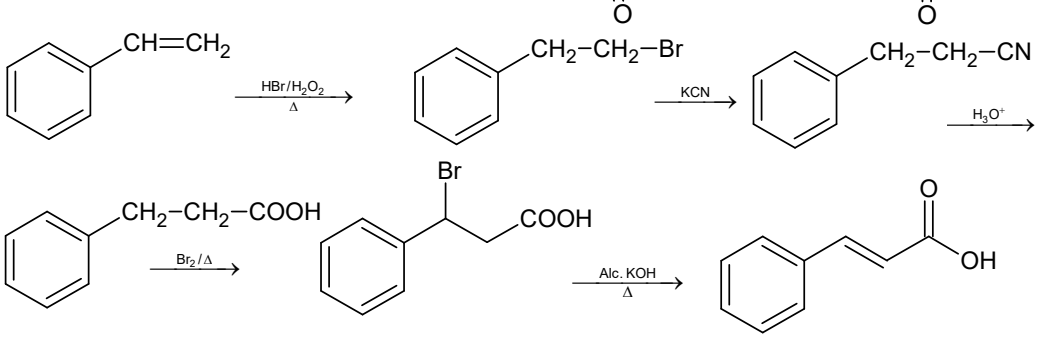
(II)



(III)



(IV)



29. D

30. C

SECTION – C

31. 00003.00

Sol. $Cu^{2+} + 2e^- \longrightarrow Cu \qquad H_2O \longrightarrow 2H^+ + \frac{1}{2}O_2 + 2e^-$

Q = 9.65 × 10 = 96.5 C

Moles of H_2SO_4 produced = moles of Cu^{2+} reduced $= \frac{96.5}{2 \times 96500} = 5.0 \times 10^{-4}$

Conc. of H_2SO_4 (M) = $\frac{5 \times 10^{-4}}{1} = 5 \times 10^{-4}$ mole / L

$\left[H^+\right] = 2 \times 5 \times 10^{-4} = 10^{-3}$ mole / L

pH = - log[H^+] = 3

32. 00002.60

Sol. $\Delta T_f = iK_f.m$

$= 2 \times 1.86 \times \frac{0.65 \times 1000}{930} = \frac{2 \times 186 \times 0.65}{93}$ = 4 × 0.65 = 2.6

33. 00082.00

Sol. $^{238}_{92}X \longrightarrow ^{234}_{90}Y \longrightarrow ^{234}_{88}Z \longrightarrow ^{234}_{89}L \xrightarrow{-7\alpha} ^{206}_{(75)}M$

7 + 75 = 82.00

34. 00001.25

Sol. $(NH_4)_2SO_4 + 2NaOH \longrightarrow 2NH_3 + Na_2SO_4 + 2H_2O$

n = 2.5 n = 5

$CuSO_4.5H_2O + 4NH_3 \longrightarrow \left[Cu(NH_3)_4\right]SO_4 + 5H_2O$

n = 2 n = 5

limiting

Hence, moles of $[Cu(NH_3)_4]SO_4$ = $\frac{5}{4} = 1.25$

35. 00006.00

Sol. $XeF_4 + H_2O \longrightarrow Xe + XeO_3 + HF + O_2$

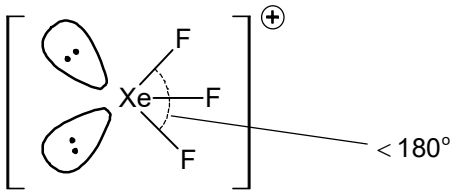
'A' 'B'

Oxidation state of Xe = 0

Oxidation of Xe in XeO_3 = +6

36. 00000.00

Sol.



Mathematics

PART – III

SECTION – A

37. A, B, C

Sol. We have $\log_{1/3}(x^2 + 2Px + P^2 + 1) \geq 0 \Rightarrow x^2 + 2Px + P^2 + 1 \leq 1$

$\Rightarrow (x + P)^2 \leq 0 \Rightarrow x = -P$. Now $x = -P$ satisfy $kx^2 + kx - k^2 \leq 0$

$\Rightarrow kP^2 - kP - k^2 \leq 0 \Rightarrow k^2 + kP - k P^2 \geq 0 \Rightarrow k^2 - k(P^2 - P) \geq 0$

Apply $D < 0$, $(P^2 - P)^2 \leq 0 \Rightarrow P^2 - P = 0 \therefore P = 0$ or 1

38. A, C

Sol. $8x^3 = 3x^2 + 3x + 1 \Rightarrow 9x^3 = x^3 + 3x^2 + 3x + 1$

$9x^3 = (x + 1)^3$, $x + 1 = \sqrt[3]{9}x$

$$x = \frac{1}{\sqrt[3]{9} - 1} = \frac{9^{2/3} + 9^{1/3} + 1}{9 - 1}$$

$$= \frac{\sqrt[3]{81} + \sqrt[3]{9} + 1}{8}$$

39. A, D

Sol. $(5x - y) + x + 2y = 2(2x + y) \Rightarrow 2x = y$

$(x - 1)^2(y + 1)^2 = (xy + 1)^2 \Rightarrow (x - 1)(y + 1) = \pm(xy + 1) \Rightarrow -x - 1 = 1$

$\Rightarrow x = -2$, $y = -4$ and $2x^2 - x - 1 = -2x^2 - 1 \Rightarrow x = \frac{1}{4}$, $y = \frac{1}{2}$

40. B, C, D

Sol. $|z|(z + 1)^8 = z^8|z + 1| \Rightarrow |z|\,|z + 1|^8 = |z|^8\,|z + 1| \Rightarrow |z + 1|^7 = |z|^7 \Rightarrow |z| = |z + 1|$

41. A, C, D

Sol. $^{100}C_{50} = \frac{100!}{50!50!}$

Exponent of 2 in 100! = 97

Exponent of 2 in 50! = 47

42. C, D

Sol. $x^2 - x + 1 = 2$, $x^2 - x - 1 = 0$

$x = \frac{1 + \sqrt{1 + 4}}{2} \Rightarrow x = \frac{\sqrt{5} + 1}{2}$, (since $x > 0$)

$x^2 - x + 1 = 1 \Rightarrow x^2 - x = 0 \Rightarrow x = 0$, $x = 1$ since $x \in (0, 2)$

43. A, D

Sol. Replace x by $\frac{1}{x}$ and differentiate and put $x = 1$

44. A, B

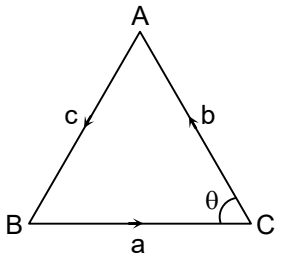
Sol. $f'(x) = \frac{2x(x^4 - 5x^2 + 4)}{2 + e^x} = 0 \Rightarrow 2x(x^2 - 4)(x^2 - 1) = 0 \Rightarrow 2x(x - 2)(x + 2)(x - 1)(x + 1) = 0$

45. B

46. D

Sol. **(for Q.45-Q.46):**

$\overline{AB} = |\bar{c}| = 2\sqrt{3}$, $AC = |\bar{b}| = 2$, $BC = |\bar{a}| = 2$, $\cos\theta = \frac{4+4-12}{2\cdot 2\cdot 2} = \frac{1}{2} \Rightarrow \theta = \frac{2\pi}{3}$



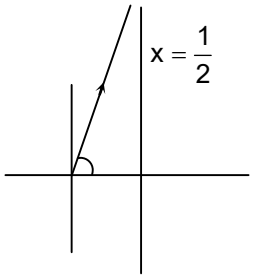
(I) (Q) $\int_a^b (f(x) - 3x)dx = a^2 - b^2 \Rightarrow \int_a^b f(x)dx = \frac{1}{2}(b^2 - a^2) \Rightarrow f(x) = x \Rightarrow f\left(\frac{\pi}{6}\right) = \frac{\pi}{6}$

(II) (P)

(III) (S) $I = \frac{\pi^3}{\ln 3}\cdot\frac{1}{\pi}(\ln|\sec\pi x + \tan\pi x|)_{7/6}^{5/6} = \frac{\pi}{\ln 3}\left(\ln\sqrt{3} - \ln\frac{1}{\sqrt{3}}\right) = \pi$

(IV) Let $u = \frac{1}{1-z} \Rightarrow \frac{u-1}{u} = z$, $|z| = 1 \Rightarrow |u - 1| = |u|$

Locus of u is $x = \frac{1}{2}$ maximum argument of u is approaches towards $\frac{\pi}{2}$



47. A

48. D

Sol. **(for Q.47-Q.48):**

(I) (S) $\text{Re}\left(\frac{2iz}{1-z^2}\right) = \left(\frac{2i}{\frac{1}{z} - z}\right) = \frac{-1}{\sin\theta}$ if $z = \cos\theta + i\sin\theta$, since $|z| = 1$

Range of $\frac{-1}{\sin\theta}$ is $(-\infty, -1] \cup [1, \infty)$

(II) (R) $R_1 \to R_1 + R_3$

$$f(\theta) = \begin{vmatrix} 0 & 0 & 2 \\ -\tan\theta & 1 & \tan\theta \\ -1 & -\tan\theta & 1 \end{vmatrix} = 2(1 + \tan^2\theta) = 2\sec^2\theta \Rightarrow \text{Range } (2, \infty)$$

(III) (T) $-1 \le \frac{8(3^{x-2})}{1-3^{2x-2}} \le 1 \Rightarrow x \in (-\infty, 0] \cup (2, \infty)$

(IV) (R) $f(x) = x^{3/2}(3x - 10)$, $x \ge 0 \Rightarrow f'(x) = \frac{3}{2}x^{1/2}(3x-10) + 3x^{3/2} = \frac{15}{2}x^{1/2}(x-2) \ge 0 \Rightarrow x \ge 2$

SECTION – C

49. 00007.00

Sol. $a_n = \dfrac{\sqrt{n^2+(n-1)^2}+\sqrt{n^2+(n+1)^2}}{n} = \dfrac{4}{\sqrt{n^2+(n+1)^2}-\sqrt{(n-1)^2+n^2}}$

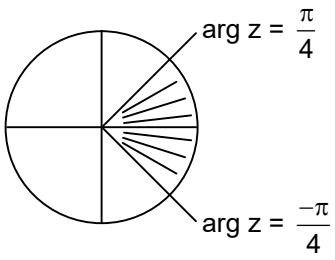
$\sum_{n=1}^{20}\dfrac{1}{a_n} = \dfrac{1}{4}\left(\sqrt{20^2+21^2}-\sqrt{1}\right) = 7$

50. 00005.00

Sol. Total sum = 4^{37}, $\left(4^{37}-2^{37}-2\right)$ = sum of coefficients divisible by 37 $\Rightarrow \dfrac{a+b+c+2}{2} = 5$

51. 00004.00

Sol. Given in the figure, $P = \dfrac{1}{4}$



52. 00016.00

Sol. f(x) – f(–x) = 6x, f(4) – f(–4) = 24 $\Rightarrow$ N = f(4) = 24 + 2286 = 2310 = 2.3.5.7.11
Number of divisors = $2^{n-1} = 16$

53. 00006.00

Sol. If tangent of P and Q meet at T, then $ST^2 = SP \times SQ \Rightarrow ST^2 = 36$, ST = 6

54. 00003.00

Sol. $f(x) = f\left(\dfrac{x}{7}\right) = \dfrac{x}{7} \Rightarrow f\left(\dfrac{x}{7^{n-1}}\right) - f\left(\dfrac{x}{7^n}\right) = \dfrac{x}{7^n} = f(x) - f\left(\dfrac{x}{7^n}\right) = \dfrac{x}{7}\left(1+\dfrac{1}{7}+\dfrac{1}{7^2}-\dfrac{1}{7^{n-1}}\right)$

Taking $x \to \infty$, $f(x) - f(0) = \dfrac{x}{6}$. Area = $\int_0^6 \left(1+\dfrac{x}{6}\right)dx = 3$